

Multiclass Classification of Alzheimer's Disease and Frontotemporal Dementia from Resting-State EEG Using Spectral Band-Power Features and Cost-Sensitive Gradient Boosting

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ARTICLE INFO	ABSTRACT
Keywords: EEG Alzheimer's disease Frontotemporal dementia Spectral band power XGBoost Machine learning Dementia classification Biomarkers	The two most common forms of early onset dementia are frontotemporal dementia (FTD) and Alzheimer's disease (AD), and their overlapping behavioral and cognitive symptoms can sometimes make clinical distinction difficult. Resting-state electroencephalography (EEG) is a low-cost non-invasive signal that captures the diffuse cortical slowing associated with neurodegeneration and has therefore been proposed as a screening biomarker to subtype dementia. This work presents a machine-learning pipeline based on relative spectral band-power features and a slowing-ratio score derived from 19-channel resting-state EEG to classify Alzheimer's disease (AD), frontotemporal dementia (FTD) and healthy-control (HC) patients. The recordings were acquired from the publicly available OpenNeuro ds004504 dataset (88 subjects: 36 AD, 23 FTD, 29 HC). The data was segmented into epochs of 4 seconds and the band-limited power in the delta, theta, alpha and beta bands was extracted per channel using Welch's method. The five features per channel were chosen as four relative band powers and a delta+theta/alpha+beta slowing ratio, resulting in a 95-dimensional feature vector per epoch. Subject-level diagnoses were obtained from a cost-sensitive XGBoost classifier, which was trained using ANOVA-F-selected features, subject-grouped five-fold cross-validation, and epoch-level soft-voting aggregation. The model achieved 69.32% subject level accuracy with cross-validation and 66.67% accuracy on a strictly held-out set of 18 subjects that were never seen during training or feature selection, with the strongest performance being for the AD and HC classes and relatively weaker recall for FTD. These results show that simple and interpretable spectral features with gradient boosting are able to significantly differentiate three overlapping clinical groups only from EEG, while emphasizing the challenge of AD-FTD confusion classification reflecting the high difficulty reported in the clinical literature. Methodological safeguards against subject-level data leakage are presented. The approach is compared with other dementia-classification studies using EEG, and the principal limitations of the study, such as sample size, single-site data collection, and epoch-count imbalance across subjects, are outlined.

1. Introduction

Dementia is a progressive clinical syndrome involving memory, language, executive function and behaviour. The two most common neurodegenerative causes of its early- and late-onset presentations are AD and FTD. The differential diagnosis between AD and FTD is of clinical importance, given the differences in prognosis, management and candidate pharmacological targets. However, the neuropsychological profiles of the two diseases often overlap and make bedside differentiation difficult, particularly in the earlier stages of disease. Definitive diagnosis typically requires a combination of neuropsychological testing, structural and functional neuroimaging, and increasingly cerebrospinal-fluid or PET-based biomarkers, all of which are expensive, time-consuming or invasive.

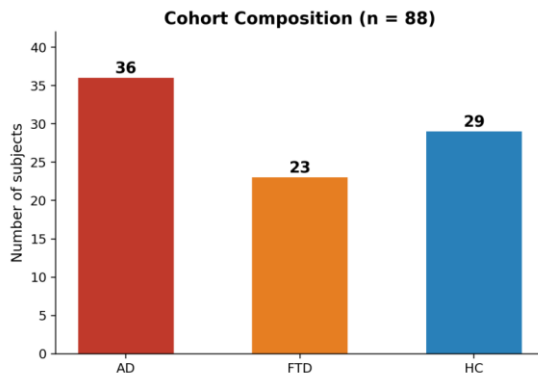


Fig. 1. Diagnostic-group composition of the study cohort ($n = 88$).

Electroencephalography (EEG) is an attractive complementary tool, as it is inexpensive, accessible, non-invasive and directly reflects the large-scale disruption of cortical rhythms associated with neurodegeneration. Previous studies have consistently reported that AD is characterised by a slowing of the EEG power spectrum, i.e. relative increases in slow-wave (delta and theta) power together with a reduction in faster alpha and beta rhythms, a pattern often summarised using delta/alpha or (delta+theta)/(alpha+beta) slowing ratios. FTD shows a partially overlapping, but overall milder and topographically restricted slowing pattern, making the three-way discrimination between AD, FTD and healthy controls (HC) considerably more difficult than any two-pairwise comparison.

We describe and evaluate an end-to-end pipeline that (i) extracts per-channel relative band-power and slowing-ratio features from short EEG epochs, (ii) trains a cost-sensitive gradient-boosted tree classifier (XGBoost) with statistical feature selection, and (iii) aggregates epoch-level predictions into subject-level diagnoses using soft voting. The pipeline is assessed with two complementary protocols: subject-grouped five-fold cross validation on the full cohort and a strictly separate 20% holdout of subjects withheld from all stages of

model fitting, including feature scaling and feature selection, to prevent subject-level data leakage. For both protocols, standard classification metrics (accuracy, per-class precision/recall/F1, and confusion matrices) are reported, and the resulting patterns of misclassification are discussed in the context of the well-documented clinical overlap between AD and FTD.

The contribution of this work is not a novel signal-processing or modelling technique, but a carefully leakage-controlled, subject-level evaluation of a compact and interpretable spectral feature set for three-class EEG dementia classification, and an honest accounting of the method's limitations. We prefer a transparent, feature-engineered approach over deep or graph-based representations as it yields an auditable model and its predictors can be traced to known electrophysiological markers of neurodegeneration.

2. Related Work

The OpenNeuro ds004504 dataset used in this study was published by Miltiadous et al. and contains 88 resting-state, eyes-closed EEG recordings from AD, FTD and HC subjects along with clinical metadata (including Mini-Mental State Examination (MMSE) scores) and both raw and pre-processed (denoised) versions of the signals in BIDS format. The same group subsequently proposed DICE-Net, a convolutional-transformer architecture for AD detection on this and related EEG data. They also reported robust classification methods and cross-validation comparisons for AD-versus-FTD-versus-control discrimination on earlier releases of the dataset.

This dataset has been used in several independent studies for benchmarking alternative feature representations and classifiers. Spectral power-ratio methods, such as theta/alpha ratios, have been investigated specifically as diagnostic markers of FTD, using a similar feature style of slowing-ratio, as applied in the present study. Connectivity-based approaches have also been investigated: a recent study used resting-state functional-connectivity metrics with a stacked ensemble of Riemannian-geometry classifiers under leave-one-subject-out cross-validation, reporting ROC-AUC values of approximately 0.82 for AD-versus-HC and 0.71 for FTD-versus-HC — generally consistent with the difficulty of FTD discrimination observed in the present results.

In general, the literature on EEG-based dementia classification ranges from classical spectral and complexity-based features with conventional classifiers (support vector machines, random forests, gradient boosting) to deep-learning approaches that operate directly on raw or minimally processed signals, and more recently, graph-based

representations that explicitly model inter-channel or inter-region relationships. Graph signal processing (GSP) methods have been applied to related EEG classification problems such as epileptic seizure detection with promising results. In GSP, the EEG montage is modeled as a graph and signals are processed in the graph-spectral domain. The current pipeline was initially scoped with a similar graph-based framing in mind; however, the implemented and evaluated methodology is based on per-channel spectral features rather than an explicit channel-graph representation, and is described as such throughout this paper. An extension to the present feature set is a formal GSP treatment of the 19-channel montage, which is a direction for future work identified in Section 6.

3. Materials

3.1 Dataset and Participants

The OpenNeuro dataset ds004504, which includes resting-state EEG recordings from 88 people who were recruited and recorded at the 2nd Department of Neurology, AHEPA General University Hospital, Thessaloniki, Greece, is used in this investigation. Three diagnostic groups comprise the cohort: 23 individuals with frontotemporal dementia (FTD group; mean age 63.6 years, SD 8.2; mean MMSE 22.17, SD 8.22), 36 individuals with a clinical diagnosis of Alzheimer's disease (AD group; mean age 66.4 years, SD 7.9; mean MMSE 17.75, SD 4.5; median disease duration 25 months, interquartile range 24–28.5).

3.2 Recording Protocol

Nineteen scalp electrodes (Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, O2) and two mastoid reference electrodes (A1, A2), sampled at 500 Hz, were used to capture the EEG using a Nihon Kohden EEG-2100 clinical system. Recordings were taken while the subjects were at rest with their eyes closed. The AD group recorded for an average of 13.5 minutes (range 5.1–21.3), the FTD group recorded for an average of 12.0 minutes (range 7.9–16.9), and the HC group recorded for an average of 13.8 minutes (range 12.5–16.5). These groups recorded for an approximate total of 485.5, 276.5, and 402.0 minutes, respectively.

3.3 Preprocessing

Instead of using the raw signals, this study makes use of the denoised derivative recordings given with ds004504. A Butterworth band-pass filter (0.5–45 Hz), re-referencing to A1–A2, Artifact Subspace Reconstruction (ASR) to suppress transient high-amplitude artifacts, and Independent Component Analysis (ICA) decomposition into 19 components with automatic rejection of components corresponding to eye and jaw movement artifacts comprised the preprocessing pipeline used by the original curators,

according to the dataset documentation. The released derivatives did not maintain the automatic annotations of leftover artifacts. These already-denoised "set" EEGLAB-format recordings found in the dataset's derivatives directory were used for all feature extraction in this investigation.

4. Methods

4.1 Channel Selection and Epoching

19-Channel 10–20 Montage Used for Feature Extraction

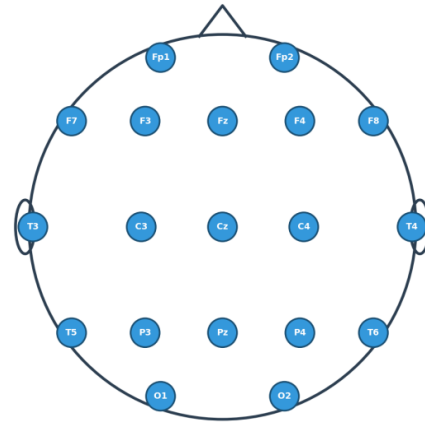


Fig. 2. The 19-channel 10–20 electrode montage used for feature extraction.

The associated EEGLAB file's 19 standard 10–20 channels were chosen for each participant (channels that weren't present were ignored), and a linear-phase FIR filter was used to band-pass filter the data between 0.5 and 30 Hz. The number of epochs per participant varied based on the total recording duration after continuous recordings were divided into non-overlapping, fixed-length 4-second epochs. Instead of analyzing entire recordings, segmenting into short epochs allows for the formation of subject-level predictions by aggregating (voting across) multiple independent estimates per subject, which tends to reduce prediction variance in comparison to a single whole-recording estimate. It also multiplies the number of training examples available to the classifier.

4.2 Spectral Feature Extraction

Welch's approach was used to measure the power spectral density (PSD) per channel over the 0.5–30 Hz range for each epoch. The four standard EEG frequency bands—delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), and beta (12–30 Hz)—were then used to calculate mean power. Two types of features were obtained for each channel: (i) the relative power in each of the four bands, which was calculated by dividing the mean power of that band by the total mean power of all

four bands for that channel and epoch; and (ii) a slowing ratio, which was calculated by dividing the sum of theta and delta power by the sum of alpha and beta power (with a small constant added to the denominator to prevent division by zero). The well-established spectral slowing linked to cortical dysfunction in dementia is condensed and clinically motivated by this slowing ratio. Each epoch is represented as a 95-dimensional feature vector with 19 channels and five features per channel (four relative band powers + one slowing ratio).

4.3 Feature Matrix and Labelling

A feature matrix of 17,419 epochs by 95 characteristics was produced by applying this process to all subjects in the AD, FTD, and HC groups. Each epoch was labeled by the diagnostic group of the subject and tagged with a subject identity that was then utilized for group-aware data splitting. As a result, subjects with longer recordings add proportionately more epochs to the pool; this asymmetry is described as a shortcoming in Section 6.3 and is not specifically adjusted for during model fitting.

4.4 Feature Scaling and Selection

A StandardScaler fitted solely on the training partition was used to standardize features (zero mean, unit variance) inside each training fold (see Section 4.6) and then applied to the matching test partition. The top 45 features were kept for model training and evaluation after a univariate ANOVA F-test (SelectKBest, scikit-learn) was fitted on the scaled training features to rank the 95 candidate features according to their relationship with the three-class diagnostic label. To ensure that no information from a test or holdout subject's epochs ever affected the scaling or selection statistics, both the scaler and the feature selector were independently re-fitted within each cross-validation fold and within the holdout split.

4.5 Classification Model

XGBoost, a regularized gradient-boosted decision-tree ensemble (Chen and Guestrin, 2016) set up for multi-class probability output (goal multi:softprob, three classes), was used for classification. With a fixed random seed (42) for reproducibility, the model employed 300 boosting rounds, a learning rate of 0.03, a maximum tree depth of 4, row subsampling of 0.8, column subsampling per tree of 0.6, an L1 regularization term (reg_alpha) of 0.1, and an L2 regularization term (reg_lambda) of 1.5. In order to prevent the minority FTD class from being consistently under-emphasized during training, per-sample training weights were calculated using scikit-learn's balanced class-weighting scheme and supplied to the classifier due to the imbalance of the three diagnostic groups (36 AD, 23 FTD, 29 HC subjects, and a correspondingly imbalanced epoch pool).

4.6 Evaluation Protocol

In order to avoid epochs from the same subject appearing in both the training and test partitions of any split—a common cause of exaggerated accuracy in epoch-based EEG classification—two complementary, subject-grouped assessment processes were employed.

Subject-grouped cross-validation, five times. In order to ensure that every epoch associated with a particular subject falls completely within either the training set or the test set, a GroupKFold split (five folds, grouped by subject identifier) was employed. Only the training subjects underwent the scaling, feature-selection, class-weighting, and model-fitting procedures outlined in Sections 4.4–4.5 inside each fold. The class with the highest mean probability was selected as the subject-level prediction after the fitted model generated per-epoch class-probability estimates for each held-out subject in that fold. These per-epoch probabilities were then averaged (soft-voted) over all of that subject's epochs. The cross-validation results shown in Section 5.1 are derived from aggregating predictions in this manner across all five folds, which results in one prediction per subject for all 88 participants in the cohort.

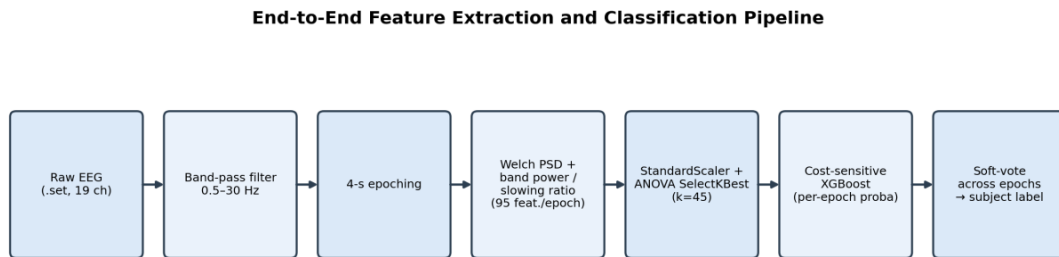


Fig. 3. End-to-end feature extraction and classification pipeline, from raw EEG to subject-level diagnosis.

Independent holdout at the subject level. A GroupShuffleSplit (test size 20%, random state 42) was used to completely set aside 18 of the 88 subjects, leaving 70 subjects for model fitting, in order to produce a single, strictly out-of-sample estimate of generalization performance. Using the identical hyperparameters and methodology as previously, the scaler, feature selector, class weights, and XGBoost model were only fitted on the epochs of the 70 training subjects; they were then applied unaltered to the epochs of the 18 held-out subjects. The cross-validation technique was followed in the formation of per-subject soft-voted predictions. This methodology offers a leakage-free assessment of performance on completely unknown patients because these 18 participants did not supply any data for scaling, feature selection, class-weight computation, or model fitting (Section 5.2). A production model, refit on the same 70 training subjects with identical hyperparameters, was additionally used to demonstrate end-to-end batch inference directly from raw EEGLAB files for a subset of the holdout subjects, illustrating the deployment path from a new patient recording to a per-class probability output.

5. Results

5.1 Subject-Level Cross-Validation Performance

The model obtained a subject-level accuracy of 69.32% for all 88 subjects under five-fold subject-grouped cross-validation with soft-voting aggregation (61 of 88 subjects correctly categorized). The aggregated confusion matrix is shown in Table 2, while per-class precision, recall, and F1-score are summarized in Table 1.

Table 1. Subject-level classification report under five-fold subject-grouped cross-validation (soft-voted across epochs).

Class	Precision	Recall	F1-score	Support
AD	0.72	0.78	0.75	36
FTD	0.67	0.43	0.53	23
HC	0.68	0.79	0.73	29
Accuracy			0.69	88
Macro avg	0.69	0.67	0.67	88
Weighted avg	0.69	0.69	0.68	88

Table 2. Subject-level confusion matrix, five-fold cross-validation (rows: true diagnosis; columns: predicted diagnosis).

True \ Predicted	AD	FTD	HC
AD (n=36)	28	1	7
FTD (n=23)	9	10	4
HC (n=29)	2	4	23

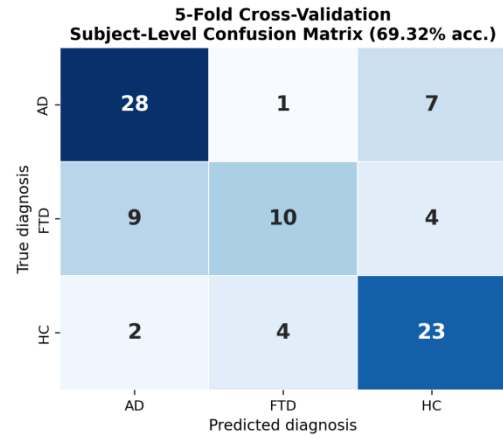


Fig. 4. Cross-validation confusion matrix heatmap (subject-level, soft-voted).

The model scored best on the AD and HC classes, correctly categorizing 28 of 36 AD patients (77.8% recall) and 23 of 29 HC participants (79.3% recall). Only 10 out of 23 FTD subjects were correctly identified (43.5% recall) on the FTD class; nine FTD subjects were incorrectly classified as AD, the single largest error category in the confusion matrix, which is consistent with the clinical and electrophysiological overlap between slowing related to AD and FTD that was discussed in Section 1.

5.2 Independent Holdout Performance

The model obtained 66.67% subject-level accuracy (12 out of 18 subjects properly categorized) on the strictly isolated 18-subject holdout set (data leakage guarded, as explained in Section 4.6). Table 3 and Table 4 report the related categorization metrics and confusion matrix.

Table 3. Subject-level classification report on the independent 18-subject holdout set.

Class	Precision	Recall	F1-score	Support
AD	0.64	0.88	0.74	8
FTD	1.00	0.38	0.55	8
HC	0.50	1.00	0.67	2
Accuracy			0.67	18
Macro avg	0.71	0.75	0.65	18
Weighted avg	0.78	0.67	0.64	18

Table 4. Confusion matrix on the independent 18-subject holdout set.

True \ Predicted	AD	FTD	HC
AD (n=8)	7	0	1
FTD (n=8)	4	3	1
HC (n=2)	0	0	2

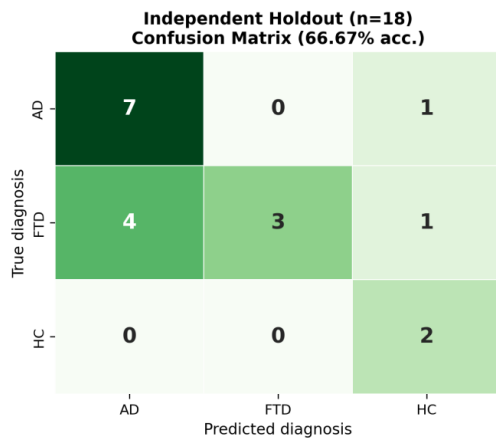


Fig. 5. Independent holdout confusion matrix heatmap (subject-level, soft-voted).

The cross-validation findings and the holdout results are often in agreement: FTD recall remained the lowest of the three classes, with four out of eight FTD participants predicted as AD, while FTD precision was flawless (no non-FTD subject was predicted as FTD). Although both HC participants in the holdout set were correctly identified and AD recall was strong (7 of 8), this figure should be viewed cautiously due to the small holdout support for the HC class

(n = 2). Although both estimates have large confidence intervals given the sample size, the similarity between the 69.32% cross-validation accuracy and the 66.67% independent holdout accuracy indicates that the model is not significantly overfitting to the cross-validation folds.

5.3 Feature Importance

Gain-based feature importances were derived from the trained XGBoost model for the 45 features that the ANOVA F-test selection kept after the production model was fitted on the 70 training subjects. The top 15 traits are displayed in Fig. 6 according to their respective relevance score. All of the top predictors are single-band relative power features, in contrast to the composite slowing-ratio feature, which is not included in the top 15. The ranking is dominated by posterior and occipito-parietal alpha power (O2, P3, O1, T5, T6, Pz, and P4 relative alpha), followed by temporal/frontal relative theta power (F7, T3, Fz) and frontal relative beta power (Fp1, Fp2). Occipital alpha power held disproportionate weight in the model compared to any other single channel-band combination, as seen by the fact that the single most significant feature, O2 relative alpha power, scored approximately 50% higher than the second-ranked feature (P3 relative alpha).

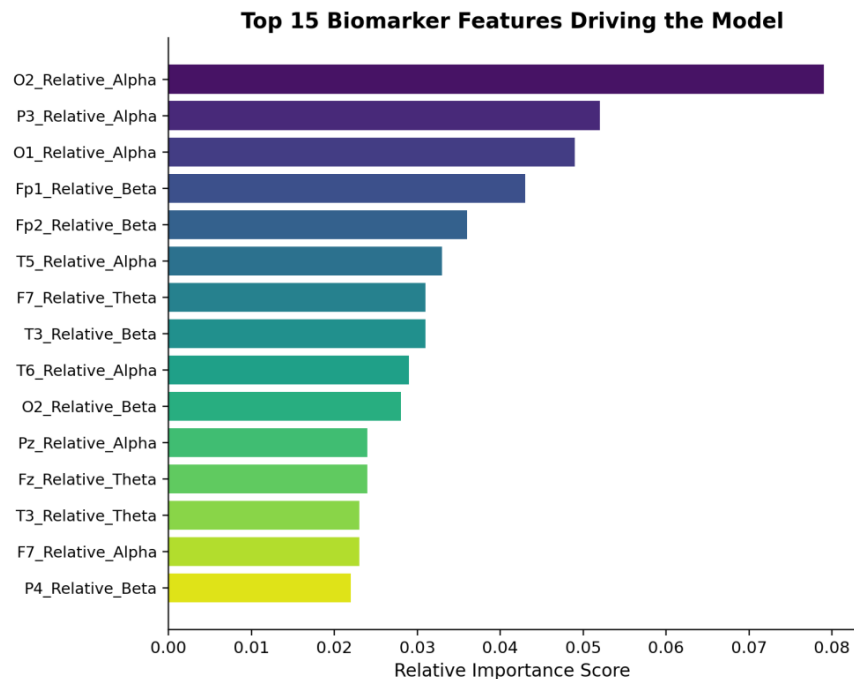


Fig. 6. Top 15 features by XGBoost gain-based importance, production model (70 training subjects).

6. Discussion

6.1 Interpretation of Results

The reported subject-level accuracies (69.32% cross-validation, 66.67% independent holdout) are generally consistent with the accuracy range reported by other groups using spectral or connectivity-based features on the same or closely related dementia EEG datasets, and they significantly surpass the 33% expected by chance in a balanced three-class problem. A recurrent finding in the EEG-dementia literature is mirrored by the consistent pattern across both evaluation protocols, where AD and HC are separated relatively well while FTD is frequently confused with AD. FTD EEG slowing tends to be milder and more spatially restricted than AD slowing, which narrows the electrophysiological margin between the two conditions relative to either group's separation from healthy controls.

An additional, model-internal perspective of this pattern is provided by the feature-importance analysis in Section 5.3. Among the top-ranked features, posterior and occipitoparietal relative alpha power (O2, P3, O1, T5, T6, Pz, P4) predominate. This is consistent with the known slowing and reduction of the posterior dominant alpha rhythm in AD, which has historically been one of the earliest and most reliable quantitative EEG markers of the illness. The supplementary contribution of frontal/temporal relative theta power (F7, T3, Fz) and frontal relative beta power (Fp1, Fp2) indicates that the model also incorporates anterior fast- and slow-wave changes that have been linked to the severity of AD and FTD, respectively. Notably, the hand-crafted delta+theta/alpha+beta slowing ratio was not among the top 15 individual predictors, despite being the feature most directly inspired by the clinical literature. This suggests that the single-band relative power terms from which it is derived carried more discriminative information in this cohort than their fixed linear combination.

6.2 Comparison with Related Work

The overall three-way classification difficulty is consistent with connectivity-based stacked-ensemble results on the same dataset, which likewise report weaker separability for FTD than for AD relative to controls. The slowing-ratio feature employed here is conceptually related to the theta/alpha and other spectral power-ratio indices that have been specifically proposed as FTD diagnostic markers. The current pipeline depends on a tiny, hand-engineered, per-channel feature set, in contrast to graph-signal-processing techniques that model explicit inter-channel structure and deep architectures like DICE-Net that learn representations directly from raw or barely processed EEG. In contrast to methods that can acquire richer spatial and temporal structure

directly from the data, this preserves the model's auditability and allows each input feature to be traced back to an established electrophysiological construct, albeit perhaps at the expense of some prediction performance.

6.3 Limitations

When evaluating these data, a number of restrictions should be taken into account. First, the small size of the underlying cohort (88 subjects overall, with only 23 in the FTD group) reduces statistical power, increases the likelihood that the particular 70/18 holdout split used here is not entirely representative of the population, and widens the confidence intervals surrounding the reported accuracies. Second, all data originate from a single clinical site (AHEPA University Hospital, Thessaloniki) using a single recording device and protocol; performance on data from other sites, devices, or population strata (for example, different age distributions, comorbidities, or dementia severities) is untested and may be lower. Third, epoch-level class weighting alone does not fully correct for this per-subject imbalance because feature vectors are computed at the level of 4-second epochs and subjects with longer recordings contribute proportionately more epochs to the training pool. Future work could investigate a subject-level or epoch-count-normalized weighting scheme. Fourth, the claimed performance depends on the quality of the dataset curators' ASR/ICA preparation pipeline because this work employs the already-denoised derivative recordings given with ds004504 rather than carrying out independent artifact rejection. Lastly, even though this was the original methodological framing for the project, the classifier and feature set described here do not implement graph signal processing over the electrode montage; readers interested in an explicit graph-based treatment of the 19-channel spatial structure should treat the current spectral-feature pipeline as a baseline rather than a GSP method.

7. Conclusion and Future Work

This study used a cost-sensitive XGBoost classifier with ANOVA-based feature selection and epoch-level soft-voting aggregation to evaluate a spectral band-power and slowing-ratio feature set for three-class EEG classification of Alzheimer's disease, frontotemporal dementia, and healthy controls at the subject level under leakage control. The method produced 66.67% accuracy on an independent, strictly isolated holdout set of 18 subjects and 69.32% subject-level accuracy under five-fold subject-grouped cross-validation on the OpenNeuro ds004504 dataset. It performed best for AD and HC and worst for FTD, which is consistent with the known clinical and electrophysiological overlap between AD and FTD. Future research could go in a number of different directions: (i) an explicit graph signal processing treatment of the 19-channel montage, modeling inter-electrode relationships rather than just per-channel spectral

content, as originally intended for this project; (ii) validation on external, multi-site EEG-dementia cohorts to assess generalizability beyond the AHEPA dataset; (iii) subject-aware or epoch-count-normalized training weights to correct for unequal per-subject epoch contributions; (iv) the addition of other feature families, such as spectral entropy, connectivity, or nonlinear complexity measures, which have demonstrated complementary value in related studies; and (v) a prospective evaluation against clinician-adjudicated

diagnoses on recently collected recordings to assess screening utility.

Appendix A. Reproducibility: Console Output Logs

This appendix replicates the raw console output produced by the pipeline described in Section 4, exactly as created by the training and evaluation code, for transparency and repeatability. The results listed in Section 5 are exactly in line with these logs.

```
... --- Step 4: Training Optimized Cost-Sensitive XGBoost Classifier ---

=====
FINAL SUBJECT-LEVEL PERFORMANCE (TUNED HIGH-ACCURACY)
=====
True Patient-Level Accuracy: 69.32%

Detailed Clinical Report:
      precision    recall  f1-score   support

      AD      0.72      0.78      0.75        36
      FTD     0.67      0.43      0.53        23
      HC      0.68      0.79      0.73        29

   accuracy                0.69        88
  macro avg              0.69      0.67      0.67        88
weighted avg              0.69      0.69      0.68        88

      AD  FTD  HC
AD   28   1   7
FTD   9  10   4
HC    2   4  23
```

Fig. A1. Console output of the five-fold subject-grouped cross-validation run (Section 5.1), showing the classification report and confusion matrix underlying Tables 1–2.

```

=====
INDEPENDENT HOLDOUT TESTING ACCURACY METRICS
=====

True Out-of-Sample Testing Accuracy: 66.67%

Holdout Test Set Classification Breakdown:
      precision    recall  f1-score   support

      AD         0.64         0.88         0.74          8
      FTD         1.00         0.38         0.55          8
      HC         0.50         1.00         0.67          2

   accuracy              0.67          18
  macro avg              0.71         0.75         0.65          18
 weighted avg              0.78         0.67         0.64          18

Confusion Matrix Layout:
      AD  FTD  HC
AD      7    0    1
FTD     4    3    1
HC      0    0    2

```

Fig. A2. Console output of the independent holdout evaluation (Section 5.2), showing the classification report and confusion matrix underlying Tables 3–4.

```

=====
TRAINING PATIENTS SET (70 subjects)
=====
sub-002, sub-003, sub-004, sub-006, sub-007, sub-008, sub-009, sub-010, sub-012, sub-014, sub-015, sub-016, sub-017, sub-018, sub-020, sub-021, sub-022
=====
HOLDOUT TESTING PATIENTS SET (18 subjects)
=====
sub-001, sub-005, sub-011, sub-013, sub-019, sub-023, sub-027, sub-034, sub-054, sub-065, sub-066, sub-068, sub-069, sub-077, sub-080, sub-081, sub-085
=====

```

Fig. A3. Console listing of the 70 training-subject and 18 holdout-subject identifiers produced by the GroupShuffleSplit, confirming that the two sets are disjoint at the subject level.

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